Communications and Networks (Spring 2022)

Problem Set 1

Issued: Feb. 28 Due: Mar. 13

Readings: Simon Haykin - Digital Communication Systems 5.1-5.10

1. Let a discrete random variable X takes values in the set $\{x_1, x_2, \cdots, x_N\}$. Prove that the entropy of X satisfies the inequality

$$H(X) \leq \log N$$

and with equality if and only if the probability $x_i = 1/N$ for all i. (Hints: review the slides)

2. A discrete memoryless source has an alphabet of 7 symbols whose probabilities of occurrence are as described here:

$$\begin{pmatrix} s_0 & s_1 & s_2 & s_3 & s_4 & s_5 & s_6 \\ 0.25 & 0.25 & 0.125 & 0.125 & 0.125 & 0.0625 & 0.0625 \end{pmatrix}$$

   a) Compute the Huffman code for the source. When combining, give a priority to the more 'combined' symbol. Compute the mean and variance of the codeword length.

   b) Compute the Huffman code for the source. When combining, give a priority to the less 'combined' symbol. Compute the mean and variance of the codeword length.

   c) Verify the entropy for the source equals to the average codeword length in a) and b). Explain why the efficiency is 100%.

   d) Compare the variance of the codeword length in a) and b) and comment on the result.

3. Prove the chain rule of entropy

$$H(X_1 X_2 \cdots X_k) = \sum_{i=1}^{k} H(X_i | X_{i-1} \cdots X_1).$$

4. For random variables of given distributions

$$X \sim \begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \end{pmatrix}, \quad Y = X^2,$$

find $H(XY),\ H(Y|X),\ H(X|Y),\ I(X;Y)$.

5. Prove the following properties of differential entropy:

a) For a random variable X and $a \neq 0$,

$$h(aX) = h(X) + \log |a|.$$

b) For two distributions $p(x), q(x)$ and a random variable X that follows $p(x)$,

$$h(X) \leq -\int p(x) \log q(x) dx$$

and with equality if and only if $p(x) = q(x)$.

c) For a random variable X, s

$$h(X) \leq \frac{1}{2} \log(2\pi e \sigma^2), \quad \text{if } \mathbb{E}(X^2) \leq \sigma^2$$

and with equality if and only if X is Gaussian. (Hints: use b))

6. (Challenge) We will investigate Gaussian channels in this question. We define the entropy of random vector $\vec{X} = \{X_1, X_2, \cdots, X_n\}$ as the joint entropy of the random variables in the vector, i.e., $H(\vec{X}) = H(X_1 X_2 \cdots X_n)$.

a) Consider a Gaussian random vector $\vec{X} \sim \mathcal{N}(\vec{\mu}, \mathbf{K}_{\vec{X}})$, show that

$$h(\vec{X}) = \frac{1}{2} \log \left[(2\pi e)^n \det(\mathbf{K}_{\vec{X}}) \right].$$

b) Assume the input is $\vec{X} \sim \mathcal{N}(\vec{\mu}, \mathbf{K}_{\vec{X}})$, the channel matrix is $\mathbf{H}$, and the addictive Gaussian noise is $\vec{Z} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$. Then the output is $\vec{Y} = \mathbf{H}\vec{X} + \vec{Z}$, show that

$$I(\vec{X}; \vec{Y}) = \frac{1}{2} \log \det \left(\mathbf{I} + \frac{\mathbf{H}\mathbf{K}_{\vec{X}}\mathbf{H}^T}{\sigma^2} \right).$$

c) Consider a parallel Gaussian channel

$$\begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix},$$

where

$$\begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \right).$$

With power constraint $\mathbb{E}(X_1^2) + \mathbb{E}(X_2^2) \leq P$, find the capacity

$$C = \max_{p_{X_1 X_2}(x_1, x_2)} I(\vec{X}; \vec{Y}).$$